measurable. Of course, this requires that we definitively find a black hole in the heavens and then construct equipment sensitive enough to detect the radiation that it emits. If the black hole were light enough, the latter step is well within the reach of current technology. Even though this experimental program has not as yet met with success, it does re-emphasize that the chasm between string theory and definitive physical statements about the natural world can be bridged. Even Sheldon Glashow—the archrival of string theory through the 1980s—has said recently, “when string theorists talk about black holes they are almost talking about observable phenomena—and that is impressive.”7

The Remaining Mysteries of Black Holes

Even with these impressive developments, there are still two central mysteries surrounding black holes. The first surrounds the impact black holes have on the concept of determinism. In the beginning of the nineteenth century the French mathematician Pierre-Simon de Laplace enunciated the strictest and most far-reaching consequence of the clockwork universe that followed from Newton’s laws of motion:

An intelligence that, at a given instant, could comprehend all the forces by which nature is animated and the respective situation of the beings that make it up, if moreover it were vast enough to submit these data to analysis, would encompass in the same formula the movements of the greatest bodies of the universe and those of the lightest atoms. For such an intelligence nothing would be uncertain, and the future, like the past, would be open to its eyes.8

In other words, if at some instant you know the positions and velocities of every particle in the universe, you can use Newton’s laws of motion to determine—at least in principle—their positions and velocities at any other prior or future time. From this perspective, any and all occurrences, from the formation of the sun to the crucifixion of Christ, to the motion of your eyes across this word, strictly follow from the precise positions and velocities of the particulate ingredients of the universe a moment after the big bang. This rigid lock-step view of the unfolding of the universe raises all sorts of perplexing philosophical dilemmas surrounding the question of free will, but its import was substantially diminished by the discovery of quantum mechanics. We have seen that Heisenberg’s uncertainty principle